Sean Luka Girgis

Python / AWS Data Engineer | ETL, SQL, PySpark, S3 & Redshift

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer and production technology professional with 20+ years of enterprise experience across Python, SQL, PySpark/Spark, AWS S3, Glue, Athena, Redshift, ETL/ELT pipelines, data validation, documentation, performance troubleshooting, production support, and stakeholder-facing reporting. Strong background building reliable data workflows, optimizing SQL and reporting layers, validating data outputs, documenting processes, and supporting regulated enterprise environments. Best aligned to roles requiring Python/AWS data engineering, ETL pipeline design, SQL optimization, S3/Redshift, Spark/PySpark, Pandas/NumPy-style transformation, documentation, Agile collaboration, and production support; TypeScript/Node.js/NestJS, TypeORM, JWT/authz, Jest/Vitest, EMR Serverless, CodeBuild, SSM, and advanced backend API ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, validation, and decision support.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable extraction, transformation, querying, analytics, and downstream consumption.
- Developed Python and SQL workflows for ETL lifecycle needs including extraction, transformation, validation, publishing, reconciliation, and production support of reporting datasets.
- Used Spark/PySpark concepts including DataFrames, Spark SQL, partition-aware design, Parquet-oriented storage patterns, and schema discipline to support scalable data processing.
- Implemented data quality and validation practices across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and correctness of outputs.
- Optimized SQL queries, Oracle schemas, Redshift workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Created runbooks, metric definitions, data flow documentation, validation checkpoints, support procedures, and stakeholder notes to improve transparency and repeatable support.
- Partnered with infrastructure, analytics, application, engineering, and business stakeholders in Agile-style delivery to troubleshoot issues, clarify requirements, and produce reliable datasets and dashboards.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Investigated real-user and synthetic monitoring source data to identify data quality issues, performance trends, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing pipelines.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by application, middleware, infrastructure, and operations teams.
- Provided troubleshooting guidance, onboarding support, training material, and knowledge sharing to improve adoption of monitoring and reporting standards.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Built validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Automated schema conversion, data validation, phased cutover support, troubleshooting, and production-readiness activities for a mission-critical migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

ServiceCall AI RAG Demo
Built staged RAG proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, retrieval decisions, Pydantic answer contract schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.
Technologies: Python, FastAPI, Docker, Pydantic, TF-IDF, Hybrid Retrieval, pytest

Production Data Quality and RCA Workflow
Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, and issue documentation.
Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

HorizonScale — Capacity Forecasting Engine
Built Python/PySpark workflows to process infrastructure telemetry, validate pipeline outputs, prepare forecasting datasets, and deliver capacity insights through repeatable reporting workflows.
Technologies: Python, Pandas, PySpark, SQL, Prophet, Data Validation